

Cephas Osei-Bonsu

+1 (704) 239-3752 | cephas.bonsuosei@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Livingstone College

Bachelor of Science in Computer Science and Mathematics

Salisbury, NC

Expected Dec 2029

- GPA: **4.0/4.0**; Presidential Scholar
- Relevant Coursework: Data Structures, Object-Oriented Programming, AI, ML, Calculus I & II

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, Java, HTML/CSS

Backend & Systems: FastAPI, Node.js/Express, PostgreSQL, Redis, Celery, REST/WebSocket APIs, JWT, Docker

Frontend & Mobile: React, Next.js, React Native, Flutter, Expo, Tailwind CSS

AI, Data & IoT: RAG, LLM APIs, OCR, Embeddings, NLP Scoring, AWS, Supabase, ESP32-C3, Paystack

TECHNICAL EXPERIENCE

Founder & Software Engineer, Vital-Go | [Live Demo](#)

2025 – Present

AI-Powered Real-Time Healthcare Monitoring Platform

- Architected IoT health platform connecting ESP32-C3 devices, FastAPI/Postgres/Redis, & Next.js dashboards
- Developed **100+** REST/WebSocket endpoints across **8** API domains with **25+** production migrations
- Implemented per-device JWT auth, secret rotation, Redis vitals cache, and **20 req/min/device** ingestion limits
- Engineered **4** Celery workers for alerts, rolling aggregates, AI risk scoring, and clinical summaries
- Validated MVP with **7** real users, zero support tickets, and **<2s** device-to-dashboard updates

Undergraduate Researcher | [GitHub](#)

Feb. 2026 – Apr. 2026

AI-Driven GPU Scheduling Optimization – UR2PhD Research Project

- Researching multi-tenant ML cluster scheduling to improve GPU utilization, queue latency, and fairness
- Built FastAPI/Next.js simulator comparing FIFO scheduling against constraint-aware job packing
- Designed live dashboards for queue depth, GPU allocation, utilization, wait time, and per-GPU job placement

Software Engineer, 404 Sentinel | [GitHub](#)

Hackathon Winner

MS-CC / ESIIL Cybersecurity and Environmental Data Hackathon

- Led 24-hour build of a winning cybersecurity platform generating explainable **0–100** sensor trust scores
- Implemented **8-layer** anomaly detection across correlation, z-score, drift, temporal, clustering, and pattern checks
- Monitored **6** Sarajevo PurpleAir sensors with reliability breakdowns, attack simulation, and blind CSV analysis

PROJECTS

Interview Prep AI | [Live Demo](#) | *FastAPI, PostgreSQL, RAG, LLM, ElevenLabs*

- Built voice-led AI interview prep for SWE, PM, cybersecurity, internships, and software roles
- Orchestrated live sessions with role-based questions, tracking, interviewer prompts, and LLM follow-ups
- Served **300+** users, reaching **78.6%** completion with **4.64** questions/session across mock interviews

SikaBoafo, Ghana Merchant OS | [GitHub](#) | *FastAPI, PostgreSQL, Redis, Flutter, Paystack*

- Building finance and inventory OS for Ghanaian microbusinesses with **43** test users and Africa-wide goals
- Architected offline-first SQLite layer with **14** tables, **16** migrations, and device-keyed sync
- Built **50-endpoint** FastAPI backend across **13** modules with PostgreSQL and Paystack infrastructure
- Delivered **6** business modules across **21** screens, plus QR, mobile money, and payment-link flows

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Founding President, ColorStack at Livingstone College

Apr. 2026 – Present

- Founded **50+** member technical community supporting mentorship, workshops, and internship readiness
- Secured **\$2K+** in early support for programming, outreach, and member growth

Full-Scholarship Scholar, HBCU Entrepreneurship Empowerment Summit

Jun. 2026

- Selected as the **only Livingstone College student** awarded a full scholarship to attend national summit
- Representing campus within **\$16.8M** PNC entrepreneurship center serving **101+** HBCUs nationwide
- Award covers registration, hotel, travel support, meals, panels, workshops, networking, and founder programming